16/01/25 MIS

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[24ESEC11]

FIRST SEMESTER B.Tech EXAMINATION, JANUARY -2025

PROGRAMMING FOR PROBLEM SOLVING – 1

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		Questions	Blooms Level	CO's	Marks
		PART-A	**5 x2=10**		
1.	**a.**	What are expressions in Python?	L1	CO1	**02**
	b.	List the standard data types in Python.	L1	CO2	**02**
	c.	Is string a mutable data type in Python? Justify your answer.	L5	CO3	**02**
	d.	How do you append data to an existing file without overwriting it?	L3	CO4	**02**
	e.	Distinguish between an abstract class and interface with example.	L3	CO5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain the Python Interactive Shell, and mention the advantages of using the Interactive Shell.	L2	CO1	**05**
		OR			
	b.	Explain how to concatenate two strings in Python? Give an example.	L2	CO1	**05**
3.	**a.**	Distinguish between a list and a tuple in Python?	L3	CO2	**05**
		OR			
	b.	How do you sort a list in Python? Compare sort() and sorted().	L3	CO2	**05**
4.	**a.**	How are strings defined in Python? What are the differences between single quotes, double quotes, and triple quotes for strings?	L3	CO3	**05**
		OR			
	b.	Explain the steps to read a file in Python. What is the role of the read() and readlines() methods?	L3	CO3	**05**
5.	**a.**	How can you create a new file in Python and write content to it?	L3	CO3	**05**
		OR			
	b.	Write a Python script to rename files with the format MM-DD-YYYY to DD-MM-YYYY.	L3	CO4	**05**
6.	**a.**	What is the purpose of assertions in Python? How are they used?	L3	CO4	**05**
		OR			
	b.	What is an exception in Python, and how do you raise one manually?	L3	CO4	**05**

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7.	a.	Write a Python program to overload + operator.	L3	CO5	**05**
		OR			
	b.	Distinguish between classes and objects in Python.	L3	CO5	**05**
		PART-C	**6 X 10=60**		
8.	a.	List all the comparison operators in Python and explain their use.	L2	CO1	**10**
		OR			
	b.	Draw a flowchart and write a program that uses a for loop to print prime numbers between 1 to 100.	L2	CO1	**10**
9.	a.	How would you use a list to implement a Magic 8 Ball program that provides random responses to user questions?	L3	CO2	**10**
		OR			
	b.	What is a dictionary in Python, and how is it different from a list? How do you add, update, and delete key-value pairs in a dictionary?	L3	CO2	**10**
10.	a.	What is the shelve module, and how is it used to save variables in Python? Give an example.	L2	CO3	**10**
		OR			
	b.	Write a program to count the number of occurrences of a specific word in a string.	L2	CO3	**10**
11.	a.	How do you write to a file in Python? Explain with an example.	L3	CO3	**10**
		OR			
	b.	What is the zipfile module, and how do you use it to create a ZIP file?	L3	CO4	**10**
12.	a.	Write a program to copy the contents of one file into another.	L2	CO4	**10**
		OR			
	b.	What are prototyping versus planning in Python? Mention their characteristics.	L2	CO5	**10**
13.	a.	Write an example of a simple Python class that models a bank account.	L4	CO5	**10**
		OR			
	b.	Demonstrate polymorphism using a common parent class with different child classes.	L4	CO5	**10**